Finance and Risk project

Part-1 Credit Risk

The dataset contains information on default payments & company details of Companies in India. You are requested to create a Credit Risk Model, using the data provided. Please use the logistic regression framework to develop the credit default model.

Importing Dataset

Data has been imported and messy columns have been corrected

Exploratory Data Analysis

Shape of data

```
data.shape
```

```
(3586, 24)
```

Description of data

```
 #   Column                                  Non-Null Count  Dtype
---  ------                                  --------------  -----
 0   Company_Code                            3586 non-null   int64
 1   Networth_Next_Year                      3586 non-null   float64
 2   Capital_Employed                        3586 non-null   float64
 3   Net_Working_Capital                     3586 non-null   float64
 4   Current_Assets                          3576 non-null   float64
 5   Total_Sales                             3586 non-null   float64
 6   Retained_Earnings                       3586 non-null   float64
 7   PBDT                                    3586 non-null   float64
 8   PBIT                                    3574 non-null   float64
 9   PBT                                     3586 non-null   float64
 10  PBT_as_perc_of_total_income             3586 non-null   float64
 11  PAT                                     3586 non-null   float64
 12  CP                                      3586 non-null   float64
 13  Book_Value_Adjusted_Unit_Curr           3569 non-null   float64
 14  EPS_Annualised_Adjustedsted_Unit_Curr   3582 non-null   float64
 15  APATM_perc                              3585 non-null   float64
 16  Creditors_Velocity_Days                 3578 non-null   float64
 17  Total_Liabilities                       3527 non-null   float64
 18  Total_Equity                            3586 non-null   float64
 19  Market_Value_of_Equity                  3586 non-null   float64
 20  total_assets                            3586 non-null   float64
 21  Cost_of_Production                      3586 non-null   float64
 22  Current_Ratio                           3585 non-null   float64
 23  Gross_Block                             3586 non-null   float64
dtypes: float64(23), int64(1)
```

Null value Check

```
data = data.fillna(data.median())
```

```
data.isnull().sum()
```

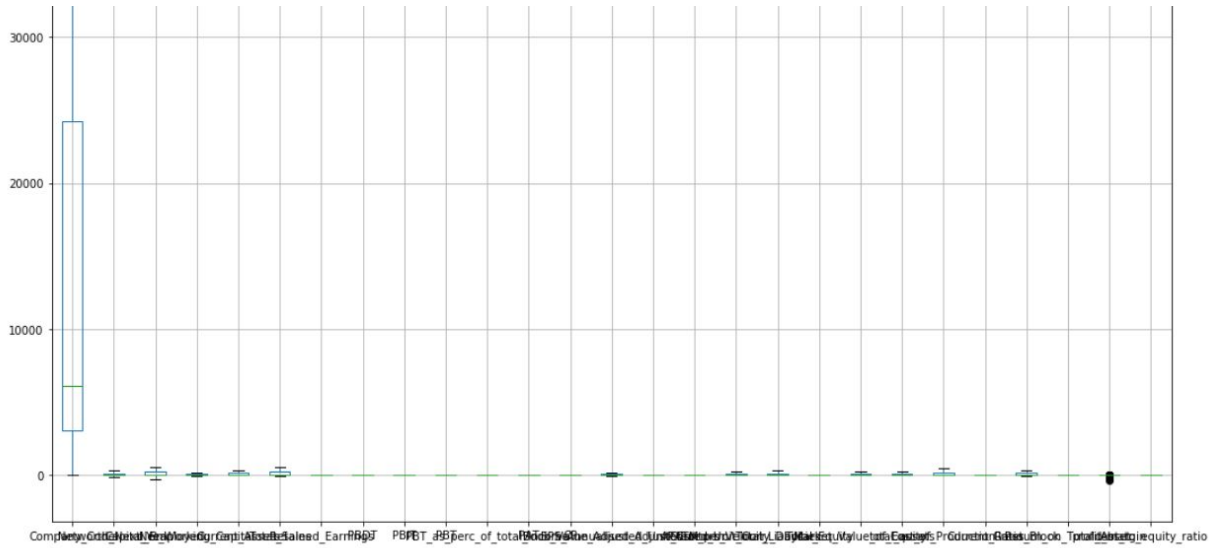
```
Company_Code                              0
Networth_Next_Year                        0
Capital_Employed                          0
Net_Working_Capital                       0
Current_Assets                            0
Total_Sales                               0
Retained_Earnings                         0
PBDT                                      0
PBIT                                      0
PBT                                       0
PBT_as_perc_of_total_income               0
PAT                                       0
CP                                        0
Book_Value_Adjusted_Unit_Curr             0
EPS_Annualised_Adjustedsted_Unit_Curr     0
APATM_perc                                0
Creditors_Velocity_Days                   0
Total_Liabilities                         0
Total_Equity                              0
Market_Value_of_Equity                    0
total_assets                              0
Cost_of_Production                        0
Current_Ratio                             0
Gross_Block                               0
dtype: int64
```

Checking Data for duplicates

```
data.duplicated().sum()
```

```
0
```

Outlier Treatment



- The dataset contains plenty of null values which have been imputed using Median
- There is only one integer type while the rest of the variables are float type
- The dataset contains 3586 observations and 24 features
- There are no duplicates in the dataset
- Outliers were identified for almost all the features which were treated by capping

New Variable Creation

Return_on_Total_Asset	profit_margin	debt_to_equity_ratio
-0.146803	-inf	14.154974
-0.065696	-0.212688	3.790509
-0.169452	-3.024227	128.984770
-0.322665	-0.591624	14.787349
-0.173831	-25.999225	114.133201

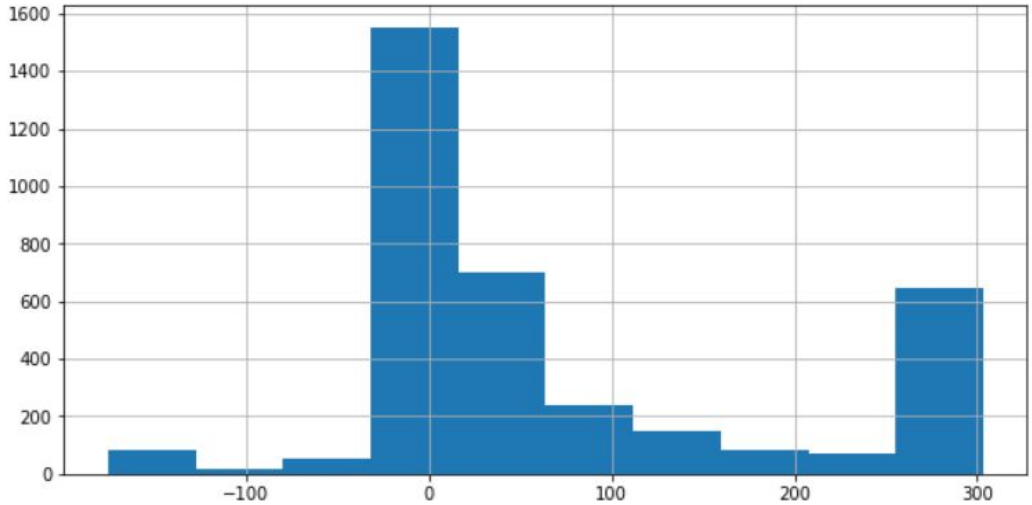
- New Variables Return on Total asset, Profit Margin and Debt to Equity ratio were created using the given formula

Target Variable Creation

- Target Variable ‘Default’ was created using Networth Next Year
- 10 % of the data shows to be defaulters

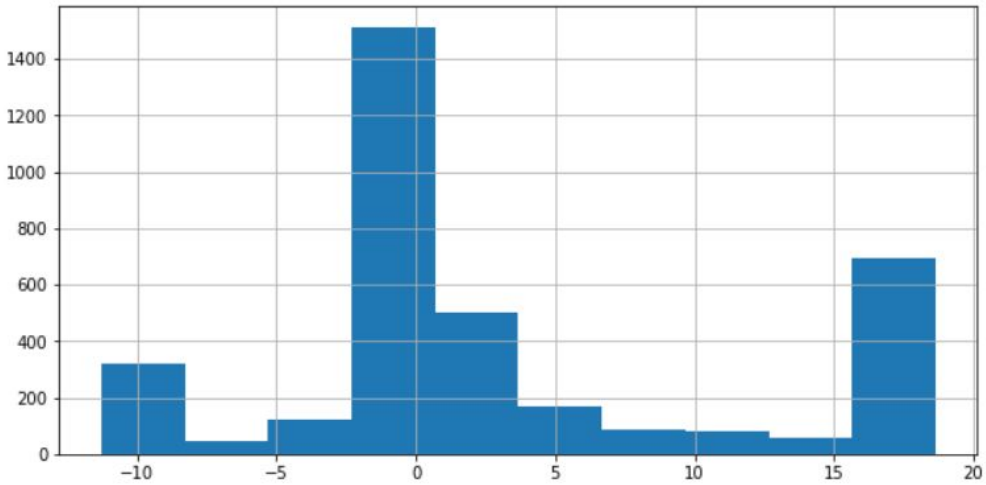
Univariate Analysis

Networth Next Year



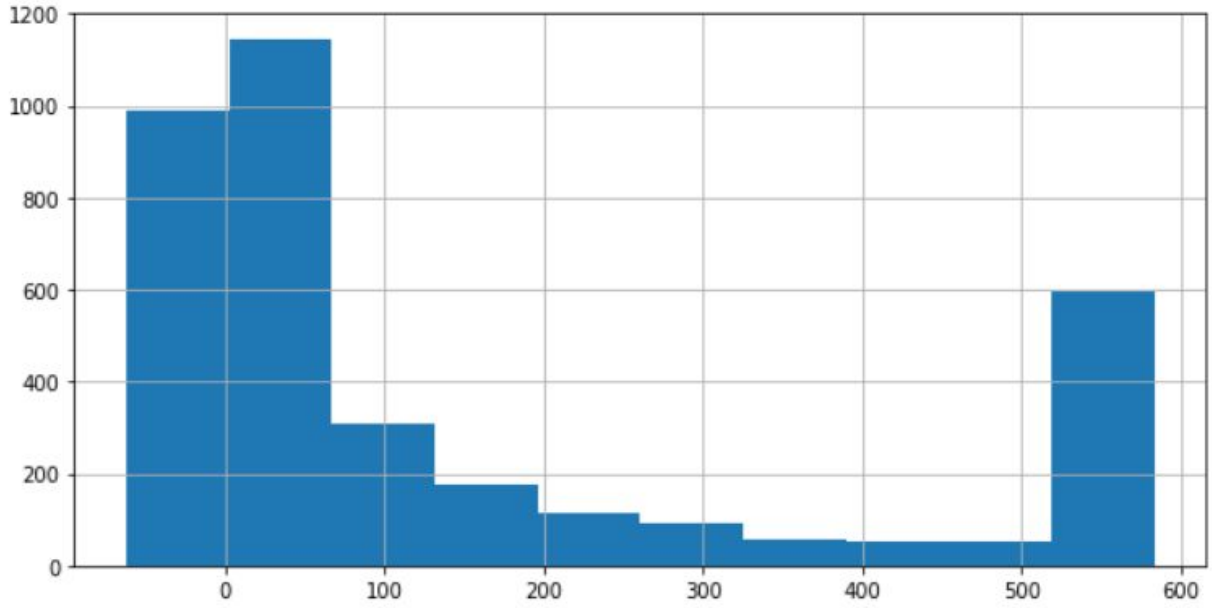
- Most of the companies are at 0 which suggests they aren't doing well

Profit Before Tax



- PBT(profit before Tax) is roughly normally distributed
- similarly, most of the companies are making close to 0 profits

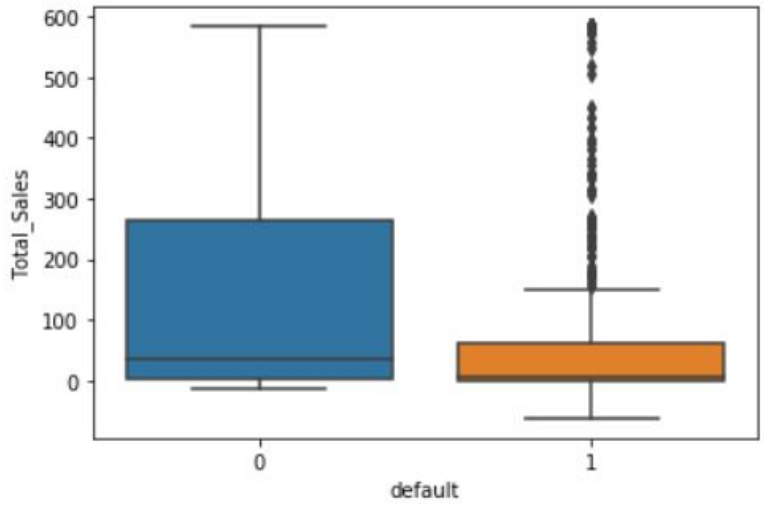
Total sales



- Total Sales is severely skewed to the right
- It suggests most of the companies are making close to 0 sales while a handful is making good Sales

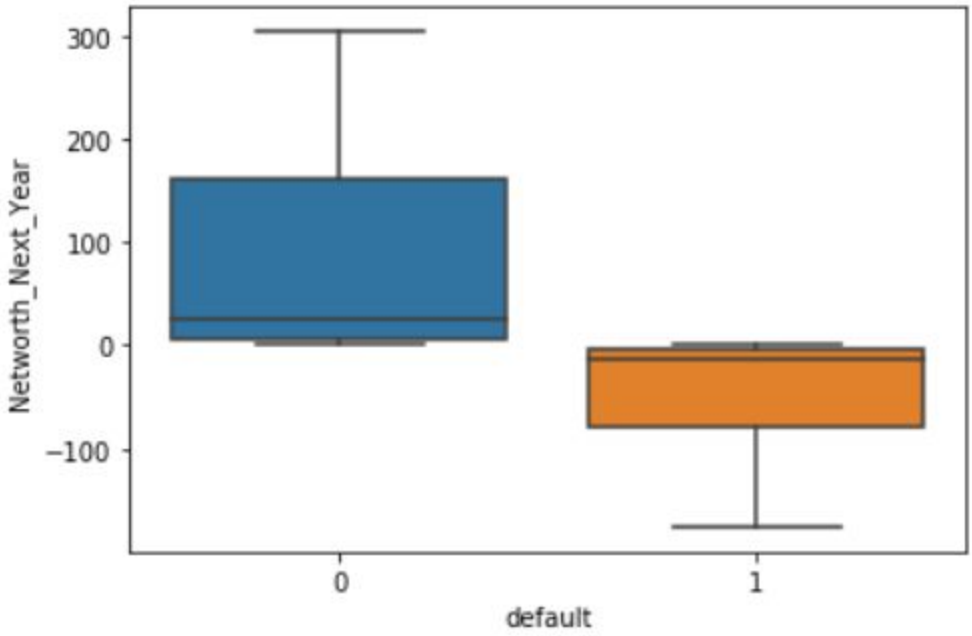
Bi-variate Analysis

Default vs Total Sales



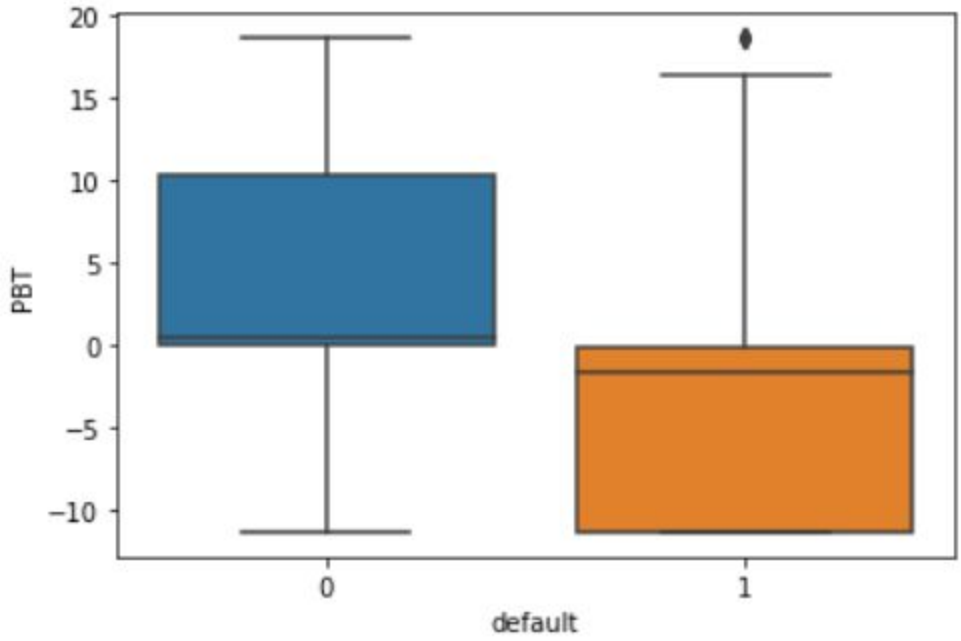
- Companies with high Sales are less likely to Default

Default vs Networth Next Year



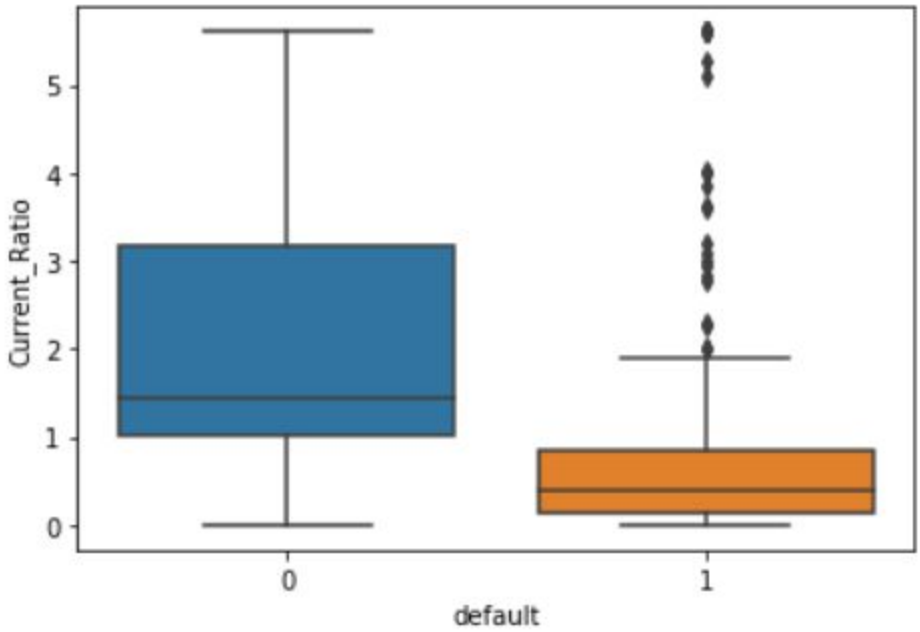
- Companies with positive Networth are not defaulting

Default vs Profit Before Tax



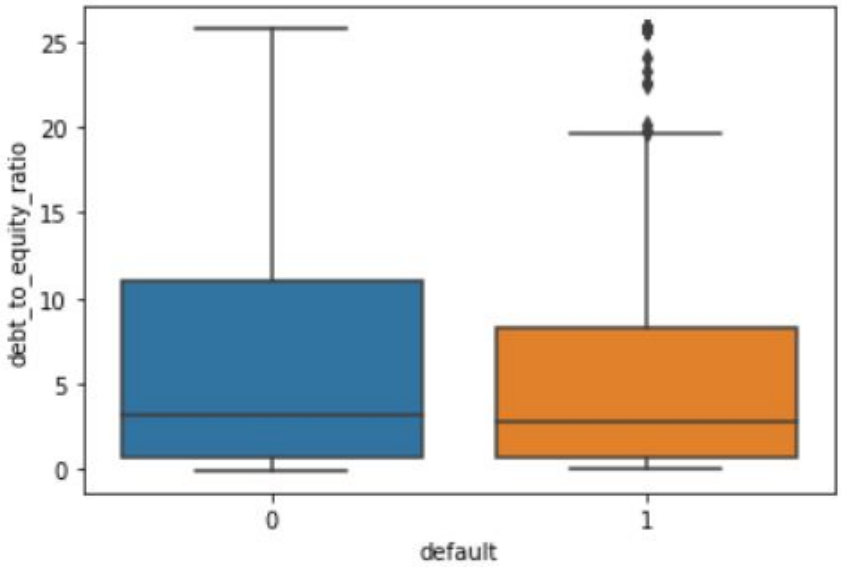
- Companies with negative profits are most likely to Default

Default vs Current Ratio



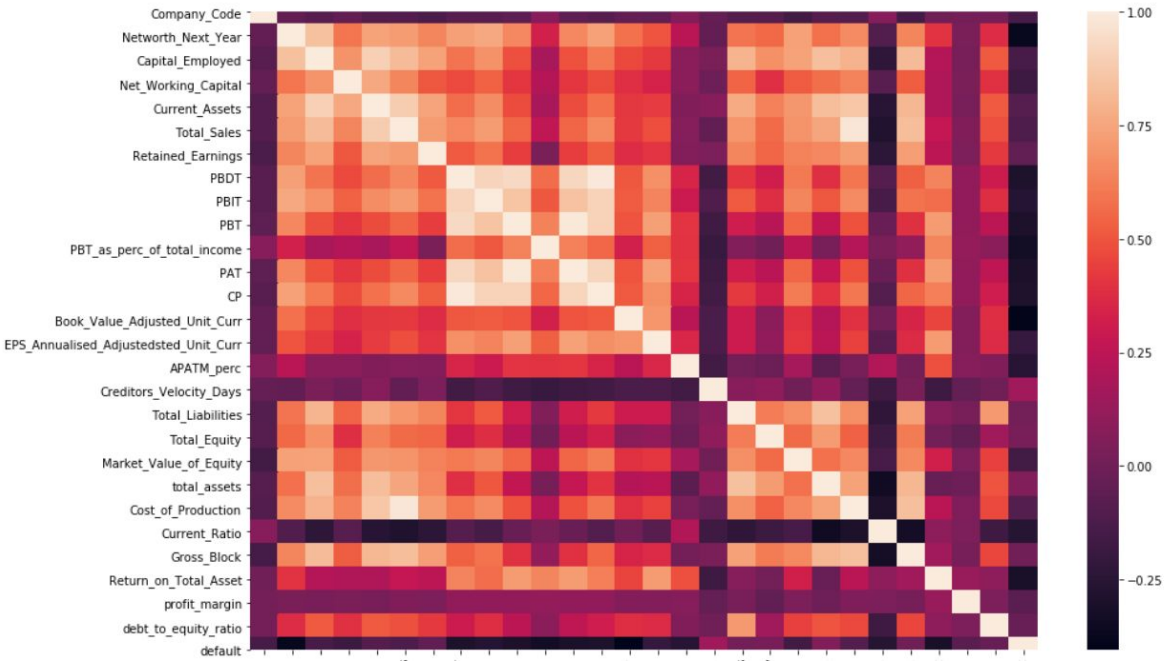
- Companies with higher Current Ratio are less likely to Default

Default vs Debt to Equity Ratio



- In the case of Debt to Equity ratio it doesn't seem to have a major effect on Default

Heatmap



- Most of the Variables are highly Correlated with each other, these variables will be ignored during Model Building to avoid Multi-Colinearity

Train-test split

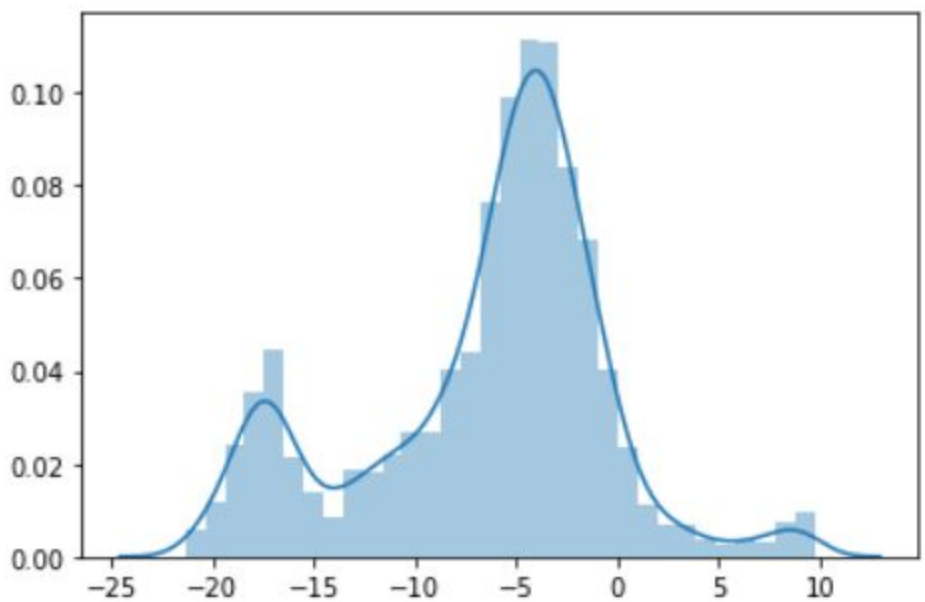
- The data has been split into train and test in an 87:33 ratio at a random state of 42

Model Building

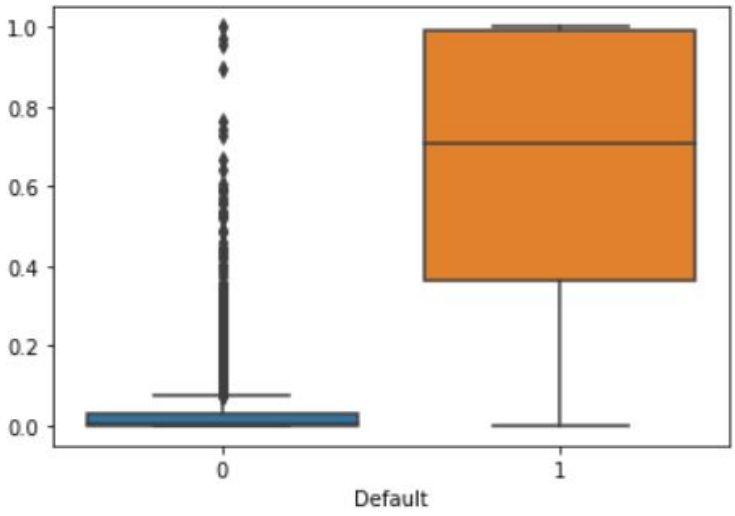
- Due to Feature engineering using Default_SMOTE Dataset the target variable has been concatenated
- After 4 iterations the model contains only significant variables

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	-0.6368	0.168	-3.792	0.000	-0.966	-0.308
PBT_as_perc_of_total_income	-0.1342	0.039	-3.399	0.001	-0.212	-0.057
PAT	-0.0687	0.035	-1.945	0.052	-0.138	0.001
Book_Value_Adjusted_Unit_Curr	-0.1171	0.010	-11.158	0.000	-0.138	-0.097
EPS_Annualised_Adjustedsted_Unit_Curr	0.1893	0.083	2.286	0.022	0.027	0.352
APATM_perc	-0.0511	0.015	-3.434	0.001	-0.080	-0.022
Current_Ratio	-0.5601	0.091	-6.154	0.000	-0.738	-0.382

Distance Plot

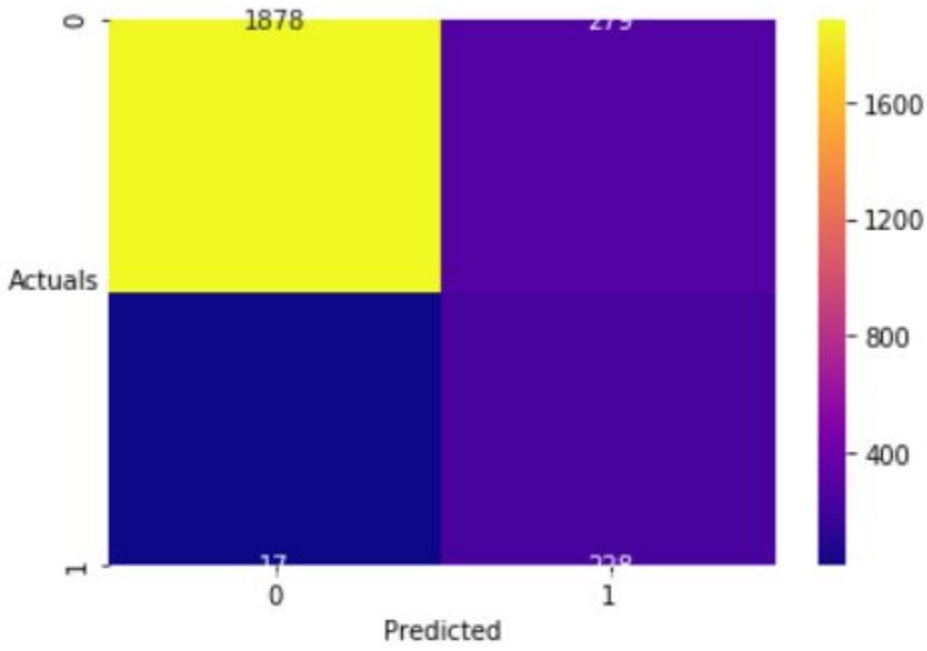


Boxplot



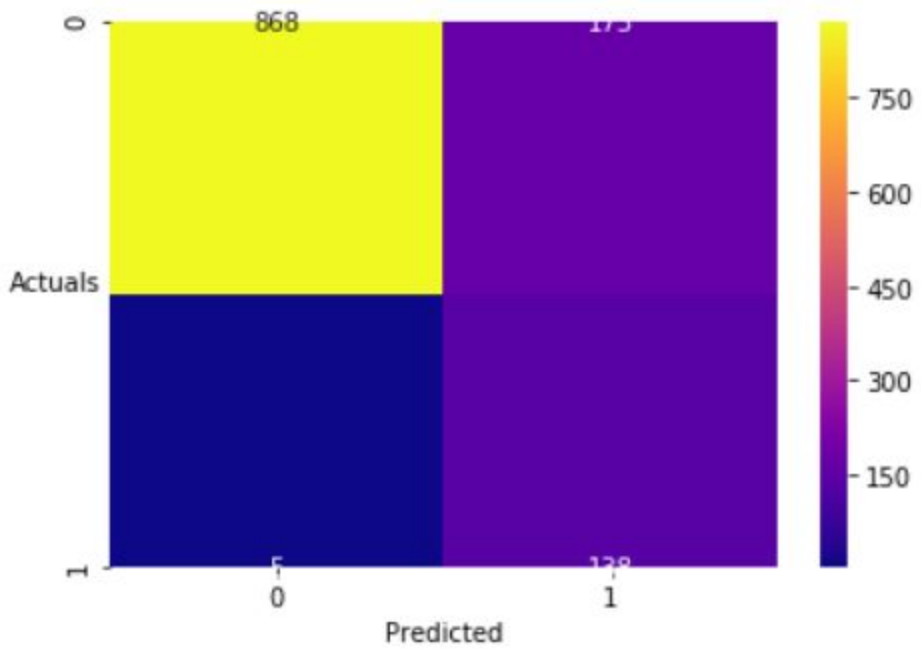
- The optimized cut-off value is taken at 0.08

Heatmap of Train Set Confusion Matrix



- Overall 88% of correct predictions to total predictions were made by the model
- Sensitivity is at 93%

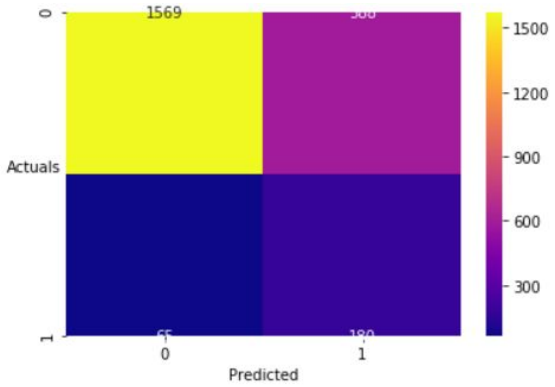
Heatmap of Test Set Confusion Matrix



- Overall 85% of correct predictions to total predictions were made by the model
- Sensitivity is at 97%
- The model evaluation shows no Overfitting or Underfitting
- The test model shows good performance

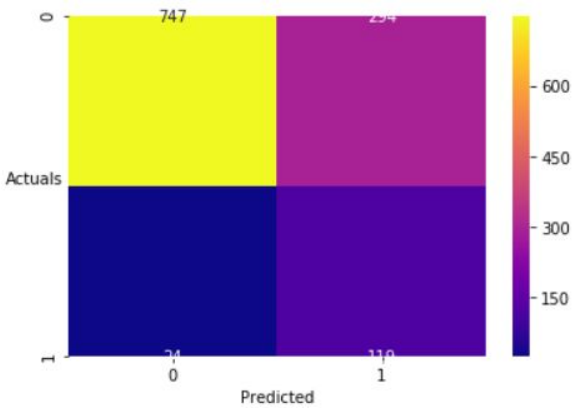
Altzman Score

Heatmap of Train Set Confusion matrix



- Overall 73% of correct predictions to total predictions were made by the model
- Sensitivity is at 73%

Heatmap of Test Set



- Overall 73% of correct predictions to total predictions were made by the model
- Sensitivity is at 83%

Comparing both Models

	LG Train	LG Test	ALTZMAN TRAIN	ALTZMAN TEST
Accuracy	0.88	0.85	0.73	0.73
Recall	0.93	0.97	0.73	0.83

- The regression model is clearly best at predicting compared to the Altzman Score

Part-2 Market Risk

Importing the Dataset

The dataset has been imported and messy columns have been fixed

Shape of the data

```
stock_prices.shape
```

```
(314, 11)
```

Summary of Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 314 entries, 0 to 313
Data columns (total 11 columns):
Date                   314 non-null object
Infosys                314 non-null int64
Indian_Hotel           314 non-null int64
Mahindra_&_Mahindra    314 non-null int64
Axis_Bank              314 non-null int64
SAIL                   314 non-null int64
Shree_Cement           314 non-null int64
Sun_Pharma             314 non-null int64
Jindal_Steel           314 non-null int64
Idea_Vodafone          314 non-null int64
Jet_Airways            314 non-null int64
dtypes: int64(10), object(1)
memory usage: 27.1+ KB
```

Checking for Null Values and Duplication

```
stock_prices.isnull().sum()
```

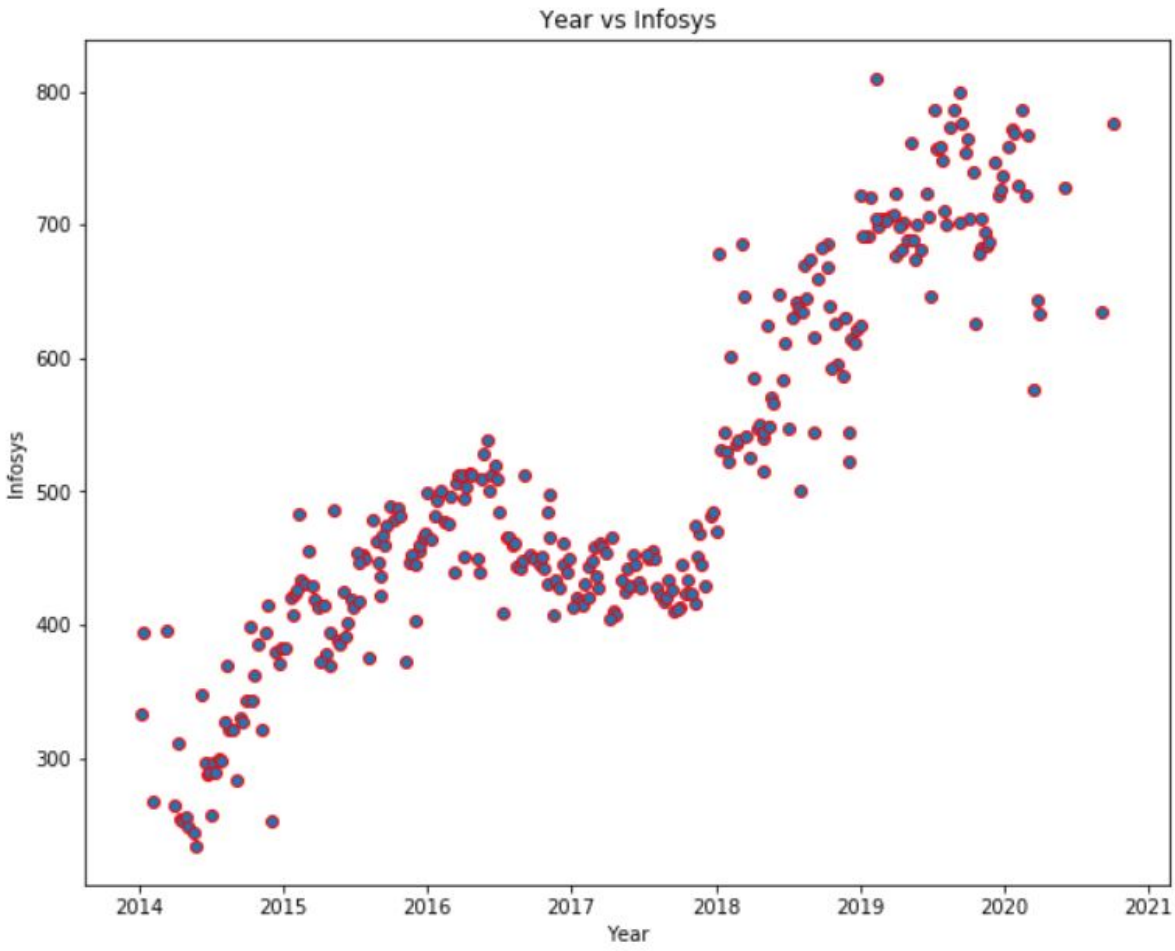
```
Date                   0
Infosys                0
Indian_Hotel           0
Mahindra_&_Mahindra    0
Axis_Bank              0
SAIL                   0
Shree_Cement           0
Sun_Pharma             0
Jindal_Steel           0
Idea_Vodafone          0
Jet_Airways            0
dtype: int64
```

```
stock_prices.duplicated().sum()
```

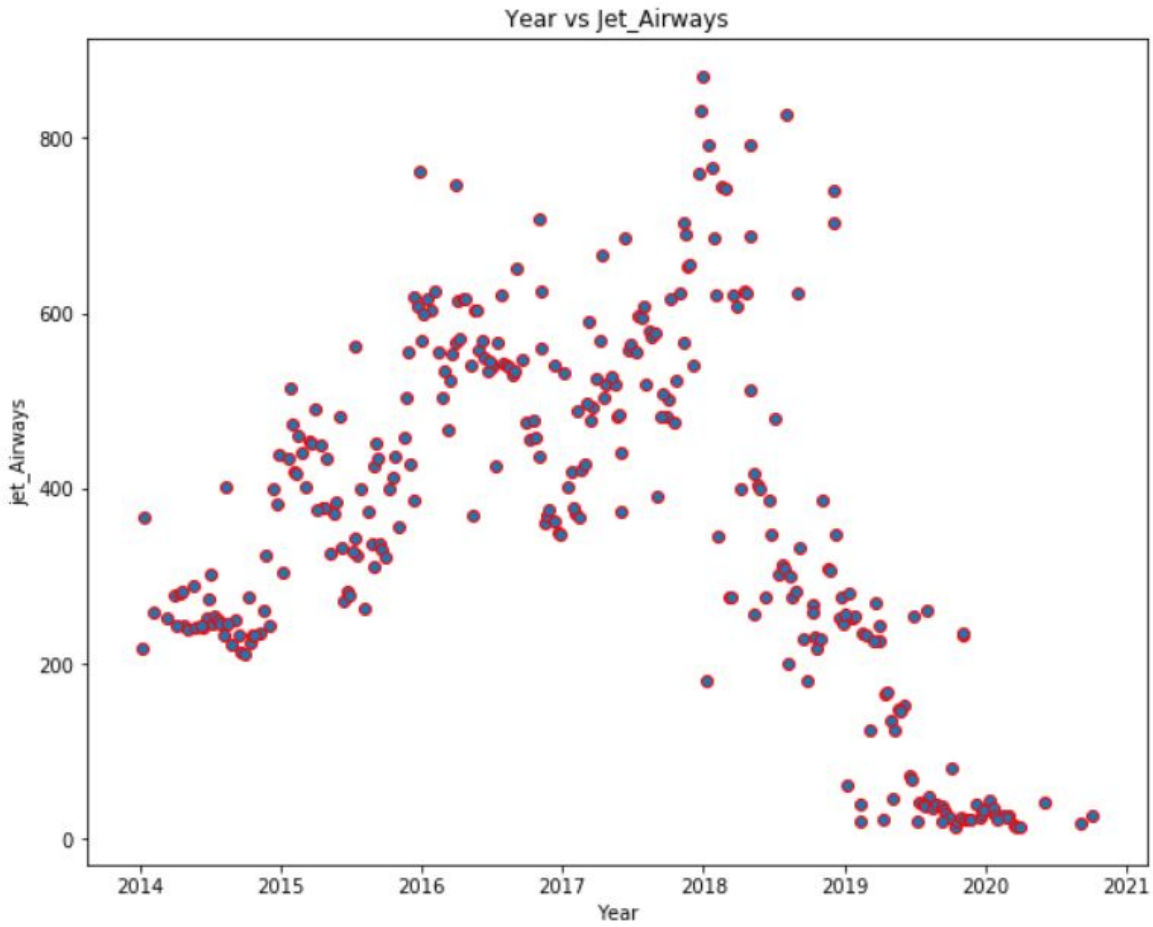
```
0
```

- There are 314 observations and 11 features
- there are no null or duplicated values in the dataset
- Except for date, every other feature is of integer type

Stock Price charts for Infosys and Jet Airways



- Over the years the Stock price of Infosys has increased higher sans a dip in 2017-18 although in 2020 there is a scatter ranging from 600 to 800



- There is no definite trend in the Stock price of Jet Airways over the years, but from 2019 the price seems to be plummeting rapidly

Stock Returns and Standard Deviation

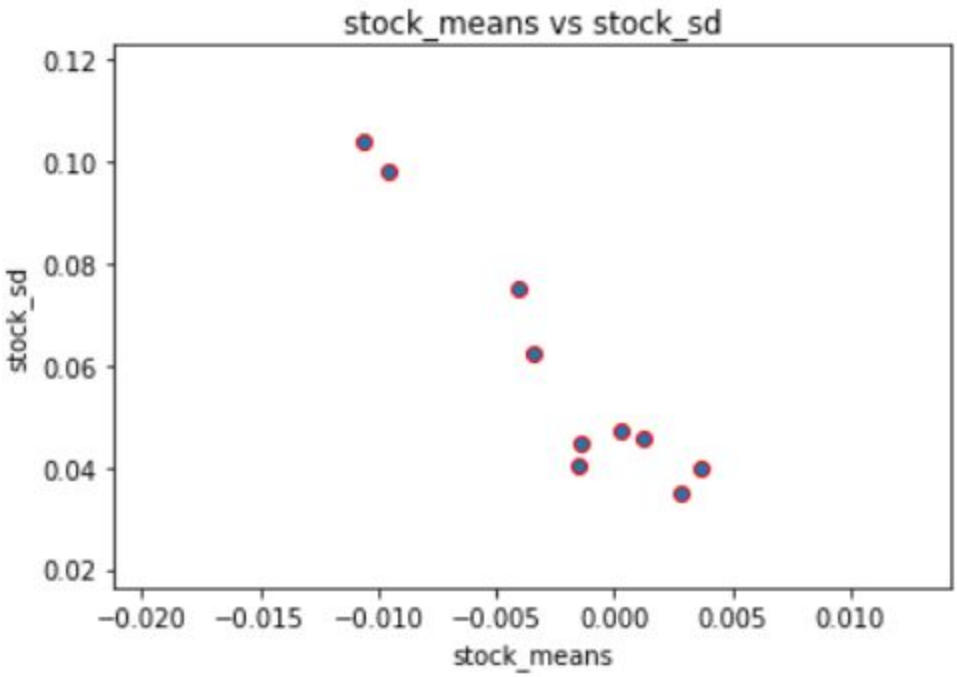
```
Infosys                 0.002794
Indian_Hotel            0.000266
Mahindra_&_Mahindra    -0.001506
Axis_Bank               0.001167
SAIL                   -0.003463
Shree_Cement            0.003681
Sun_Pharma             -0.001455
Jindal_Steel           -0.004123
Idea_Vodafone          -0.010608
Jet_Airways            -0.009548
dtype: float64
```

- Shree Cement has the highest Stock return followed by Infosys and Axis Bank

```
Infosys                 0.035070
Indian_Hotel            0.047131
Mahindra_&_Mahindra     0.040169
Axis_Bank               0.045828
SAIL                    0.062188
Shree_Cement            0.039917
Sun_Pharma              0.045033
Jindal_Steel            0.075108
Idea_Vodafone           0.104315
Jet_Airways             0.097972
dtype: float64
```

- Infosys has the lowest Risk followed by Shree Cement and Mahindra

Plotting Stock Means vs Standard Deviation



- The Scatter Plot suggests higher the return lower the risk

Insights

- Infosys and Shree Cement are both Low Risk and have high returns
- Companies with lower returns are so good as long as risk is not negative